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Prepartum nutrient intake alters glucogenic capacity of liver slices from periparturient dairy cows

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ABSTRACT

Ruminants rely on hepatic gluconeogenesis to support whole-body glucose metabolism and to supply glucose for lactose synthesis. Understanding the effect of plane of nutrition before parturition on the capacity for hepatic gluconeogenesis in dairy cows may provide a basis for improved cow health and productivity in the subsequent lactation. Our objectives were to determine the effects of far-off (FO) dry period diet, close-up (CU) period diet, and their interaction on adaptations in metabolism of gluconeogenic substrates (Ala and propionate) in liver slices. Multiparous Holstein cows ($n = 71$) were fed different amounts of nutrients during the FO and CU dry periods in a 3 (FO diet) $\times$ 2 (CU diet) factorial arrangement. During the FO period (d -60 to -25 before calving) cows received a control diet fed for ad libitum intake to meet NRC recommendations (100NRC), a moderate energy diet fed ad libitum to exceed NRC recommendation for NEL by at least 50% (150NRC), or the moderate energy diet fed at restricted intake to meet 80% of NEL requirements (80NRC). During the CU period (d -24 until parturition), cows were fed a diet either for ad libitum intake (CUA) to meet or exceed NRC recommendations or in restricted (CUR) amounts to provide 80% of calculated NEL requirements. All cows received the same lactation diet postpartum. Liver slices from biopsies at d -30 , -14 , 1, 14, and 28 relative to parturition were used to determine in vitro conversion of [$1-^{14}\text{C}$] Ala and [$1-^{14}\text{C}$] propionate to glucose and CO_2 . Gluconeogenic rates from propionate were 71%, 37%, and 73% greater than from Ala on d 1 for cows fed 100NRC, 150NRC, and 80NRC, respectively. Gluconeogenesis from propionate was significantly greater on d -14 and d 28 for cows fed 80NRC compared with 100NRC and 150NRC. Oxidation of propionate to CO_2 increased from d -14

to d 1 postpartum for all far-off treatments. On d -14 , oxidation of propionate to CO_2 tended to be lower for 100NRC versus 150NRC and 80NRC. Gluconeogenesis from Ala on d -30 tended to be less for 150NRC compared with 80NRC. Gluconeogenesis from Ala tended to be greater for CUR on d 14. Oxidation of Ala to CO_2 was not significantly different among treatments tended to be higher on d 1 postpartum for 100NRC. Close-up diets had minimal effects on gluconeogenesis and substrate oxidation. Cows fed restricted amounts of energy in the FO dry period had greater hepatic gluconeogenic capacity both prepartum and postpartum compared with those fed excess amounts of energy relative to requirements. Controlling energy intake during the far-off dry period can alter hepatic capacity for gluconeogenesis from propionate and Ala, which may confer an advantage to the cow by adapting metabolic processes before parturition and resulting in improved health and productivity.

Key words: liver, gluconeogenesis, metabolism

INTRODUCTION

Due to pregastric fermentation of dietary carbohydrates and thus, the catabolism of most dietary glucose, ruminants rely on hepatic gluconeogenesis to meet whole-body requirements for glucose (Aschenbach et al., 2010). Ruminant liver must continually synthesize glucose to meet requirements for maintenance and milk synthesis (Young, 1977). Concentrations of glucose in blood must be maintained within a narrow range to maintain normal body functions. Synthesis of glucose by the ruminant liver is under tight metabolic control and is regulated according to requirements (Aschenbach et al., 2010). Increases in glucose demand, such as that observed in periparturient dairy cattle, are typically matched by increased endogenous gluconeogenesis, predominantly in the liver (Reynolds et al., 1994; Drackley et al., 2001).

The process of gluconeogenesis is essential to meet the metabolic glucose requirements. If gluconeogenesis is compromised, results can be detrimental to cow health and productivity. Hepatic glucogenic rates are

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-25. Nonstandard abbreviations are available in the Notes.

controlled by substrate availability, quantities of rate-limiting enzymes, and the degree of activation of rate-limiting enzymes, which are influenced by hormones (She et al., 1999). In the ruminant, propionate is the main substrate for gluconeogenesis and accounts for 50% to 60% of total glucose entry in fed animals (Lomax and Baird 1983; Larsen and Kristensen, 2013). In vitro hepatic synthesis of glucose from [1-¹⁴C] propionate was greater in early lactation than in mid-lactation (Aiello et al., 1984). Amino acids (principally Ala), glycerol, and lactate all contribute carbon for gluconeogenesis (Larsen and Kristensen, 2013).

Negative energy balance during early lactation is characterized by carbohydrate shortage and subsequently increased fatty acid mobilization from adipose tissue. Periods of elevated adipose tissue catabolism, such as the time surrounding parturition, result in increased nonesterified fatty acids (NEFA) uptake into the liver and subsequent deposition of triglycerides (TG). Increased hepatic TG is associated with decreased milk production and increased incidence of periparturient disease (Arshad and Santos, 2022). Accumulation of TG in hepatocytes may constrain their ability to perform normal functions. Triglyceride accumulation in the liver diminishes hepatic function and may be involved in the development of numerous metabolic disorders (Herd, 1988; Bruss, 1993; Arshad and Santos, 2022). Hepatic TG accumulation may reduce the capacity of the liver to detoxify ammonia to urea (Strang et al., 1998), potentially leading to decreased capacity for gluconeogenesis from propionate (Overton et al., 1999). Gluconeogenesis from [2-¹⁴C] propionate was depressed by 24% in cells infiltrated with lipid due to previous exposure to 2 mM oleate for 45 h (Cadórniga-Valiño et al., 1997). The activities of 2 key enzymes in the gluconeogenic pathway, phosphoenolpyruvate carboxykinase and fructose 1,6-bisphosphatase, were reduced postpartum in cows that were overfed during the dry period to induce fatty liver (Rukkamsuk et al., 1999b).

Drackley et al. (2001) pointed out that because increases in liver mass during the periparturient period are modest, daily metabolic activity per gram of liver tissue must approximately double, based on data from Reynolds et al. (2003). Evaluating the ability of liver slices to metabolize propionate and Ala would give an indication of the effect of diet on metabolic and enzymatic capacity of liver tissue to perform oxidation and gluconeogenesis. Measuring the difference between rates of CO₂ and glucose production gives rates of oxidation of [1-¹⁴C] propionate in the TCA cycle, and the ratio of the 2 rates gives a measure of the efficiency of propionate utilization and gluconeogenesis (Knapp et al., 1992). The objective of this experiment was to determine the effect of far-off and close-up diets, varying in energy density and amount fed, on liver slice ca-

capacity for gluconeogenesis and oxidation of propionate and Ala in vitro. Our hypothesis was that cows fed a low-energy diet formulated to meet NRC recommendation in the far-off period, followed by increased feeding 3 wk prepartum, will offer the best chance to enhance gluconeogenic capacity in liver slices postpartum.

MATERIALS AND METHODS

All procedures were conducted under protocols approved by the University of Illinois Institutional Animal Care and Use Committee. The experiment from which liver samples were obtained for this study has been described by Dann et al. (2006). Multiparous Holstein cows were fed one of 2 far-off period (FO) diets (Table 1) from dry-off to -25 d before expected parturition. The control diet (100NRC) was formulated to meet but not greatly exceed NRC (2001) recommendations for all nutrients during the far-off dry period when consumed at predicted ad libitum DMI. The higher nutrient density diet exceeded NRC (2001) recommendations for nutrients during the far-off dry period and was fed for either ad libitum intake (150NRC) or in restricted amounts (80NRC). Intake was restricted to 80% of calculated NE_L requirement of the cow based on BW at dry-off (NRC, 2001). The higher nutrient density diet was similar in composition to a typical close-up diet, but did not contain anionic salts.

At -24 d before expected parturition until parturition, half of the cows from each far-off diet were randomized to be fed a close-up (CU) diet (Table 1) at either ad libitum intake (CUA) or in restricted amounts (CUR) to provide 80% of calculated NEL requirement of the cow, based on BW at dry-off (NRC, 2001). The close-up diet was similar to the far-off higher nutrient density diet except for the addition of anionic salts. Thus, treatments were in a 3 × 2 factorial arrangement of 3 far-off period treatments (100NRC, n = 25; 150NRC, n = 25; 80NRC, n = 24) and 2 close-up period treatments (CUA, n = 38; CUR, n = 36), resulting in 6 dry period treatment groups (100NRC, ad libitum close-up, n = 13; 100NRC, restricted close-up, n = 12; 150NRC, ad libitum close-up, n = 13; 150NRC, restricted close-up, n = 12; 80NRC, ad libitum close-up, n = 12; and 80NRC, restricted close-up, n = 12). After parturition, all cows were fed a lactation diet (Table 1). All diets were mixed once daily and fed as TMR. Approximately half of each diet was fed in the morning (1100 h) and the remainder fed in the late afternoon (1700 h). Alfalfa hay (~2 kg DM daily) was top-dressed on the lactation diet until 14 DIM.

Cows were housed in tiestalls throughout the experiment and were allowed to exercise daily in an outside lot for 3 h (0700 to 1000 h). Two weeks before expected parturition, cows were moved to individual box stalls until parturition.

Table 1. Ingredient and chemical composition of diets fed to multiparous Holstein cows during the dry period (from Dann et al., 2006)

Ingredient	Diet % of DM			
	100NRC ¹	150NRC, 80NRC ²	Close-up	Lactation ³
Alfalfa silage	41.68	26.24	25.14	20.10
Alfalfa hay	—	13.97	13.40	—
Corn silage	21.10	25.52	24.49	28.13
Wheat straw	26.15	—	—	—
Cottonseed	—	4.39	4.18	9.65
Ground shelled corn	7.20	17.24	16.31	25.72
Soybean meal	3.05	—	—	5.15
Expeller soybean meal	—	1.73	1.62	5.94
Soybean hulls	—	10.24	9.67	1.48
Calcium chloride	—	—	1.10	—
Limestone	—	—	1.29	1.58
Sodium bicarbonate	—	—	—	1.08
Magnesium oxide	—	—	—	0.13
Magnesium sulfate	—	—	1.81	—
Dicalcium phosphate	0.09	—	0.18	0.56
Sodium chloride	0.24	0.18	0.19	0.22
Mineral-vitamin mix ⁴	0.27	0.27	0.28	0.22
Vitamin E ⁵	0.18	0.18	0.28	0.04
Vitamin A and D ⁶	0.04	0.04	0.06	—
Chemical ⁷	—	—	—	—
n	13	12	14	15
CP	15.8 ± 0.1 ⁸	16.5 ± 0.2	15.7 ± 0.1	18.1 ± 0.2
ADF	31.8 ± 0.3	26.1 ± 0.4	25.6 ± 0.4	19.0 ± 0.3
NDF	46.5 ± 0.4	38.1 ± 0.6	36.9 ± 0.5	28.7 ± 0.4
Starch	13.2 ± 0.1	21.4 ± 0.2	20.9 ± 0.4	27.8 ± 0.3
Total fatty acids	1.7 ± 0.03	3.1 ± 0.05	2.9 ± 0.05	4.6 ± 0.1
NEL, Mcal/kg	1.30 ± 0.01	1.59 ± 0.01	1.61 ± 0.01	1.77 ± 0.01
Ash	10.9 ± 0.2	8.4 ± 0.1	11.2 ± 0.2	9.6 ± 0.2
Ca	0.92 ± 0.02	0.84 ± 0.02	1.85 ± 0.10	1.43 ± 0.03
P	0.27 ± 0.01	0.29 ± 0.01	0.37 ± 0.06	0.47 ± 0.01
Mg	0.21 ± 0.01	0.26 ± 0.02	0.45 ± 0.03	0.31 ± 0.01
K	2.16 ± 0.05	1.82 ± 0.04	2.12 ± 0.34	1.55 ± 0.02
Na	0.13 ± 0.01	0.12 ± 0.01	0.14 ± 0.01	0.52 ± 0.03

¹Far-off dry period control diet (100NRC).²Far-off dry period higher nutrient density diet fed to 150NRC and 80NRC groups.³Alfalfa hay (~2 kg DM) was top-dressed on the lactation diet from parturition to 14 DIM.⁴Contained a minimum of 5% Mg, 10% S, 7.5%K, 2.0%Fe, 3.0% Zn, 3.0% Mn, 5,000 mg/kg Cu, 240 mg/kg I, 40 mg/kg Co, 150 mg/kg Se, 2,200 kIU/kg vitamin A, 660 kIU/kg vitamin D3, 7700 IU/kg vitamin E.⁵Contained 44,000 IU/kg.⁶Contained 10,000 kIU/kg vitamin A and 1,249 kIU/kg vitamin D.⁷Calculated from monthly analysis of individual ingredients in diets.⁸Mean ± SE.

After parturition, cows were returned to tiestalls. Cows were milked twice daily (0300 and 1500 h).

Liver was sampled via puncture biopsy with a Hughes-type biopsy instrument (Hughes, 1962; Veenhuizen et al., 1991) from cows under local anesthesia at ~0700 h on d -30 ± 2, -14 ± 2, +1, +14 ± 1, and +28 ± 2 relative to parturition. Biopsies (~1 g) were placed in ice-cold PBS (0.9% NaCl) at pH 7.4 and transported to the laboratory where liver slices were prepared using a Krumdieck tissue slicer (Alabama Research and Development, Munford, AL) equipped with a 5-mm diameter tissue holder. Slices (~20 mg) were blotted and weighed into 25-mL incubation flasks for a total of ~40 mg of tissue per flask. To determine oxidative and glucogenic capacity from

L-Ala and propionate, liver slices from individual cows were incubated separately in media containing either L-Ala (10 mM; MP Biomedicals) plus [1-¹⁴C]- L-Ala (1 μCi; MP Biomedicals) or Na-propionate (10 mM; American Radiolabeled Chemicals, St. Louis, MO) plus [1-¹⁴C] propionate (1 μCi; American Radiolabeled Chemicals). Bicarbonate-free Krebs-Ringer bicarbonate medium was prepared according to procedures of Overton et al. (1999) with modifications (Piepenbrink and Overton, 2003); each flask contained 3 mL of incubation medium. After tissue loading was complete, flasks containing liver were gassed with 95:5 O₂:CO₂, a CO₂ well containing a piece of folded filter paper (#1 Whatman filter paper, 42.5 mm) was suspended in the flask using nylon monofilament,

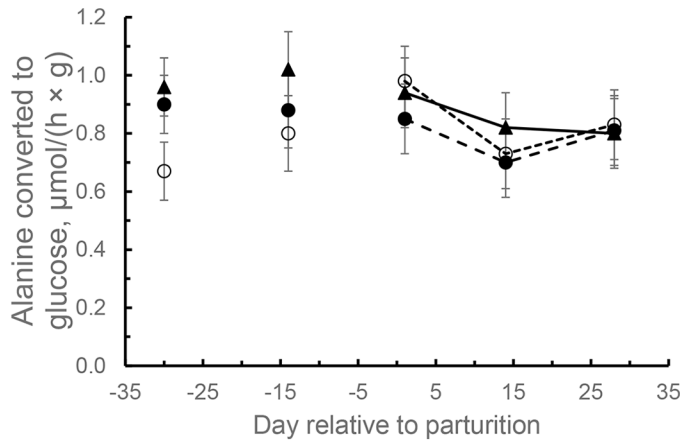


Figure 1. Least squares means and SE for conversion of L-Ala to glucose by liver slices from multiparous Holstein cows fed diets to meet 100% (100NRC; black circles and dashed line), 150% (150NRC; white circles and dotted line), or 80% (80NRC; black triangles and solid line) of NRC requirements for NEL during the far-off dry period. There was a tendency ($P = 0.11$) for higher conversion of Ala to glucose on d -30 for both 100NRC and 80NRC when compared with 150NRC. Postpartum, the effect of day was significant ($P = 0.04$) but the far-off treatment by day interaction did not reach significance ($P = 0.18$).

and the flask was sealed with a rubber septum. Flasks were incubated in a shaking water bath at 37°C for 2 h. All incubations were performed in triplicate.

To determine the rate of conversion of L-Ala or Na-propionate to CO₂ in flasks containing live tissue, incubations were stopped after 2 h by injecting 0.5 mL of 25% TCA (wt/vol) into the incubation media and 0.1 mL of 30% NaOH was injected directly onto the folded filter paper located in the CO₂ well. Blank flasks were processed identically except these flasks did not receive tissue and were stopped immediately. After incubations were stopped, live and blank flasks were shaken on ice for 1 h to collect CO₂ in the hanging well. The CO₂ wells were processed according to procedures of Drackley et al. (1991) and radioactivity in CO₂ was determined by liquid scintillation spectroscopy (Beckman LS 6000 IC, Beckman Instruments, Inc. Fullerton, CA).

After removal of rubber septa and hanging wells from the flasks, incubation media were processed according to procedures of Piepenbrink and Overton (2003). [³H] L-glucose (Sigma Chemical Co.) was added as an internal standard to account for losses of radioactivity during processing. The 3-column ion-exchange procedure of Mills et al. (1981) was used to isolate [1-¹⁴C] glucose in processed incubation media. Ion-exchange resins (Sigma Chemical Co., St. Louis, MO) were used as described by Mills et al. (1981). Radioactivity (¹⁴C and ³H) was quantified by dual-label liquid scintillation spectroscopy (model LS6000IC, Beckman Coulter, Inc.).

Table 2. Least squares means and SE for effects of close-up treatment on rates of Ala metabolism to CO₂ and glucose by liver slices

Day and product	Close-up treatment ¹			<i>P</i> -value ²
	CUA	CUR	SEM	
	μmol/(g of wet tissue × h)			
d -14				
CO ₂	0.52	0.47	0.08	0.65
Glucose	0.98	0.83	0.1	0.29
d 1–28				
CO ₂	0.60	0.58	0.01	0.46
Glucose	0.78	0.91	0.1	0.28

¹CUA = close-up dry period diet with ad libitum intake; CUR = close-up dry period diet with restricted intake.

²Probability for effect of close-up treatment. Postpartum, the interactions of close-up treatment and day for CO₂ ($P = 0.18$) and glucose ($P = 0.77$) were not significant.

Substrate conversion to end products was calculated using the known amount of radioactivity present in each flask relative to the amount of radioactivity measured in end products following the 2-h incubation period. Substrate conversion rates were corrected for tissue weight and radioactivity in blank flasks.

Cow was the experimental unit. Statistical computations were performed using SAS (SAS Institute Inc., Cary, NC, Release 9.4). Data were analyzed with linear mixed models using the MIXED procedure of SAS (Littell et al., 1996). Data from the far-off period (-30 d relative to parturition), close-up period (-14 d relative to parturition), and postpartum period (1, 14, and 28 relative to parturition) were analyzed separately as described in Dann et al. (2006) and Litherland et al. (2011). Because diets differed during the far-off dry period and close-up dry period, and postpartum data represented residual effects of dry period diets, we chose not to include all prepartum and postpartum data in a single repeated-measures analysis; this approach was also used for the full dataset (Dann et al., 2006) and by Litherland et al. (2011).

Data for the far-off period were analyzed as a random design. The model for far-off period contained the effects of far-off treatment (100NRC, 150NRC, and 80NRC). Cow nested within far-off treatment was designated as a random effect. Far-off treatment effects were separated by use of the PDIF statement when $P < 0.05$.

Data for the close-up period were analyzed as a randomized design with a 3 × 2 factorial arrangement of treatments (far-off treatments × close-up treatments). The model for close-up period data contained the effects of far-off treatment (100NRC, 150NRC, and 80NRC), close-up treatment (CUA and CUR), and the interaction of far-

Table 3. Least squares means and SE of effects of close-up treatment on propionate metabolism by liver slices to CO₂ and glucose

Day and product	Close-up treatment ¹			<i>P</i> -value
	CUA	CUR	SEM	
	μmol/(g of wet tissue × h)			
d −14				
CO ₂	3.0	3.3	0.3	0.33
Glucose	1.4	1.4	0.2	0.99
d 1–28				
CO ₂	4.1	4.2	0.2	0.54
Glucose	1.8	1.6	0.2	0.36

¹CUA = close-up dry period diet with ad libitum intake; CUR = close-up dry period diet with restricted intake.

²Probability for effect of close-up treatment. Postpartum, the interactions of close-up treatment and day for CO₂ ($P = 0.26$) and glucose ($P = 0.68$) were not significant.

off and close-up treatments. Cow nested within far-off and close-up treatment was designated as a random effect.

Data for the postpartum period were analyzed as a randomized design with a 3×2 factorial arrangement of treatments with repeated measurements. The model contained the effects of far-off treatment (100NRC, 150NRC, and 80NRC), close-up treatment (CUA and CUR), the interaction of far-off and close-up treatments, day of biopsy, and all interactions of treatments and day. Cow nested within far-off and close-up treatment was designated as a random effect. Several covariance structures were tested and the ARH(1) option yielded the lowest value of the Akaike information criterion and was used for all variables with repeated measures. Significance of the F -test of fixed effects was declared at $P < 0.05$ and trends discussed when $0.05 < P < 0.15$.

RESULTS AND DISCUSSION

Our objective was to determine the effect of far-off and close-up diets, varying in energy density and amount fed, on in vitro hepatic capacity for gluconeogenesis and oxidation of propionate and Ala across the transition period. Substrates (propionate and Ala) tested in this experiment were selected because their carbon enters the glucogenic pathways at different points and gluconeogenesis from each thus is controlled by different enzymes. Propionate enters the gluconeogenic pathway at succinyl-CoA and its contribution of carbon to glucose is regulated primarily by phosphoenolpyruvate carboxykinase (PEPCK). Because ruminants are in a constant state of gluconeogenesis, and propionate is typically the main substrate for gluconeogenesis, flux of carbon from propionate to glucose occurs constantly in fed ruminants and is subject to regulation by endocrine control and nutritional status. Alanine supplies carbon for gluconeogenesis after

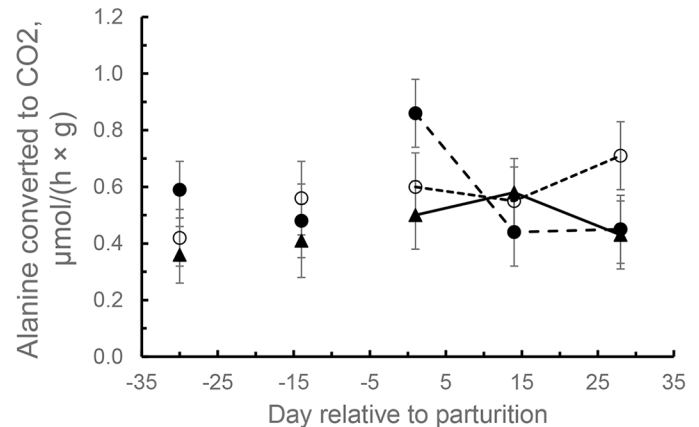


Figure 2. Least squares means and SE for oxidation of L-Ala to CO₂ by liver slices from multiparous Holstein cows fed diets to meet 100% (100NRC; black circles and dashed line), 150% (150NRC; white circles and dotted line), or 80% (80NRC; black triangles and solid line) of NRC requirements for NEL during the far-off dry period. Postpartum, the effect of time was not significant ($P = 0.24$), but there was a far-off treatment by day interaction ($P = 0.03$).

transamination and conversion to pyruvate. Pyruvate is converted to oxaloacetate by pyruvate carboxylase (PC), which therefore regulates conversion of Ala to glucose.

Close-up treatment minimally affected substrate oxidation or conversion to glucose by liver slices in this experiment (Tables 2 and 3), and interactions of close-up treatment with far-off treatment and with day of biopsy likewise were not significant; thus, discussion will focus mainly on effects of far-off treatments. Gluconeogenesis from L-Ala tended ($P = 0.11$) to be lower on d -30 for 150NRC cows compared with 80NRC and 100NRC (Figure 1). Cows fed 150NRC might have had lower capacity to perform gluconeogenesis due to consumption of the higher-energy diet and subsequent suppression of enzymes in the glucogenic pathway. Cows overfed energy during the dry period had lower activities of PC and PEPCK 1 wk prepartum (Rukkwamsuk et al., 1999b; Murondoti et al., 2004). Capacity to use L-Ala for gluconeogenesis postpartum was not different among far-off or close-up treatments.

Oxidation of L-Ala to CO₂ by liver slices on d -30 was numerically higher from cows fed 100NRC than from cows fed 150NRC or 80NRC (Figure 2). Additionally, oxidation of L-Ala by liver slices from cows fed 100NRC peaked on d 1 and tended ($P = 0.08$) to be greater than that from cows fed 150NRC or 80NRC (far-off treatment by day interaction, $P = 0.03$). Oxidation of L-Ala to CO₂ by liver slices from 150NRC-fed cows tended ($P = 0.08$) to be greater on d 28 compared with cows fed 80NRC. The flux of [1-¹⁴C] Ala through pyruvate dehydrogenase (via pyruvate) results in oxidative decarboxylation of the radiolabel and release as ¹⁴CO₂. If [1-¹⁴C] Ala is

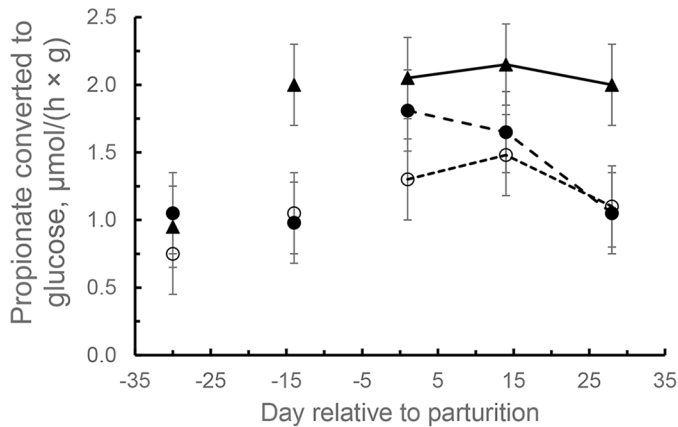


Figure 3. Least squares means and SE for gluconeogenesis from propionate by liver slices from multiparous Holstein cows fed diets to meet 100% (100NRC; black circles and dashed line), 150% (150NRC; white circles and dotted line), or 80% (80NRC; black triangles and solid line) of NRC requirements for NEL during the far-off dry period. Conversion of propionate to glucose was significantly greater on d -14 ($P < 0.05$) and d 28 ($P < 0.05$) for cows fed 80NRC then 100NRC or 150NRC. Postpartum, the effect of day tended ($P = 0.08$) to be significant, whereas there was a far-off treatment by day interaction ($P = 0.02$).

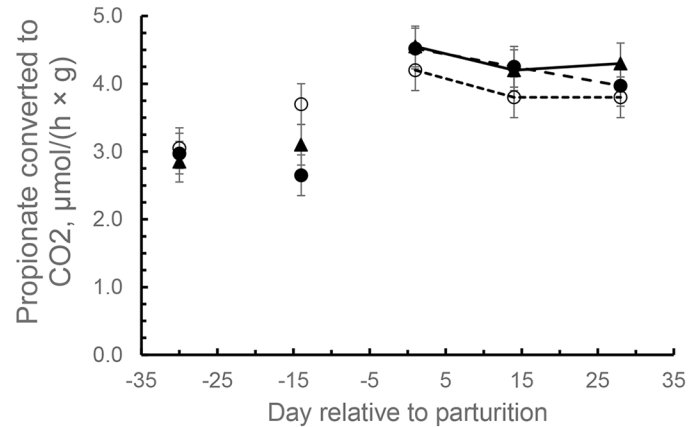


Figure 4. Least squares means and SE for oxidation of propionate to CO_2 by liver slices from multiparous Holstein cows fed diets to meet 100% (100NRC; black circles and dashed line), 150% (150NRC; white circles and dotted line), or 80% (80NRC; black triangles and solid line) of NRC requirements for NEL during the far-off dry period. On d -14 there was a trend ($P = 0.09$) for a higher rate of oxidation of propionate to CO_2 for 150NRC cows compared with that for 100NRC cows. Postpartum, effects of day ($P = 0.25$) and the far-off treatment by day interaction ($P = 0.52$) were not significant.

converted to oxaloacetate through pyruvate and PC, the radiolabel either proceeds through PEPCK or is released as $^{14}\text{CO}_2$ if it proceeds through citrate and the tricarboxylic acid cycle. Flux from pyruvate to glucose occurs through a substantially different pathway to oxaloacetate compared with that of propionate. Velez and Donkin (2005) determined that conversion of lactate to glucose was higher in feed-restricted cows in combination with greater PC mRNA expression. Additionally, Knapp et al. (1992) reported that the majority of radiolabeled Ala was converted to lactate and pyruvate by liver slices obtained from lactating dairy cows.

Gluconeogenesis from propionate by liver slices was similar for all far-off treatments on d -30 (Figure 3). However, gluconeogenesis from propionate by liver slices from 80NRC-fed cows showed a distinctly different pattern from that for cows fed 100NRC or 150NRC. Gluconeogenesis from propionate by liver slices from 80NRC-fed cows was higher ($P < 0.05$) than 100NRC and 150NRC on d -14, remained numerically higher on d 1 and 14, and was higher ($P < 0.05$) again on d 28 (Figure 3). On d 1 postpartum, gluconeogenesis from propionate by liver slices was highest for 80NRC, intermediate for 100NRC, and lowest for 150NRC. Prepartum feed restriction may have resulted in upregulation of enzymatic pathways for gluconeogenesis from propionate that persisted throughout the periparturient period and remained higher than the other far-off treatments through 28 d postpartum. Similar to our data, conversion of propionate to glucose by liver slices from feed-restricted cows in mid-lactation increased numeri-

cally (but not significantly) when compared with that of controls (Velez and Donkin, 2005).

Oxidation of propionate to CO_2 by liver slices increased on d 1 postpartum compared with -30 and -14 (Figure 4). Rates of conversion of propionate to CO_2 were similar among treatments on d -30. On d -14, however, there was a trend ($P = 0.09$) for a higher rate of oxidation of propionate to CO_2 for 150NRC cows compared with that for 100NRC cows. On d 1, 14, and 28 postpartum, the rate of oxidation of propionate to CO_2 was numerically lower for 150NRC cows than for 100NRC or 80NRC, although the effects of far-off treatment and the far-off treatment by day interactions were not significant ($P > 0.20$). Close-up treatment had essentially no effect on rates of conversion of propionate to CO_2 and glucose by liver slices (Table 3).

Liver slices from cows in this experiment exhibited greater capacity to convert either Ala or propionate to glucose on d 14 or 28 postpartum than at either d -30 or -14 prepartum. Because we employed a concentration of propionate and Ala (10 mM) that should have yielded estimates of maximal gluconeogenic capacity in ruminant liver (Knapp et al., 1992), increases in rate of glucose produced are indicative of increased cellular expression or concentration of limiting enzymes for conversion of substrate to glucose. Our data are consistent with the concept that adaptations occur in bovine liver that enable increased efficiency of utilization of substrates for gluconeogenesis when glucose demand is increased after parturition. The small increase in capacity for conversion of $[1-^{14}\text{C}]$ propionate to glucose may be a

Table 4. Least squares means and SE for the ratio of conversion of propionate to glucose with the conversion to CO₂ by liver slices from multiparous Holstein cows given different far-off and close-up diets

Day	Far-off treatment ¹				Close-up treatment ²			<i>P</i> -value ³		
	100NRC	150NRC	80NRC	SE	CUA	CUR	SE	FO	CU	FO × CU
d -30	0.39	0.28	0.37	0.07	—	—	—	0.48	—	—
d -14	0.42	0.32	0.36	0.07	0.36	0.38	0.05	0.48	0.74	0.71
d 1	0.40	0.39	0.37	0.04	0.41	0.37	0.04	0.86	0.44	0.64
d 14	0.39	0.32	0.36	0.05	0.39	0.32	0.04	0.55	0.18	0.25
d 28	0.29	0.32	0.43	0.07	0.34	0.36	0.06	0.20	0.76	0.59

¹100NRC = far-off dry period control diet with ad libitum intake; 150NRC = far-off dry period higher nutrient density diet with ad libitum intake; 80NRC = far-off dry period higher nutrient density diet with restricted intake.

²CUA = close-up dry period diet with ad libitum intake; CUR = close-up dry period diet with restricted intake.

³FO = far-off dry period effect; CU = close-up dry period effect. Interactions of far-off treatment by day ($P = 0.03$) and close-up treatment by day ($P = 0.05$) were significant. The interaction of far-off treatment by close-up treatment by day was not significant ($P = 0.43$).

consequence of milk yield, and thus demand for glucose, increasing more rapidly than DMI and, presumably, propionate supply after parturition. Propionate supply and the capacity of the liver to convert propionate to glucose may be related directly (Drackley et al., 2001). Velez and Donkin (2005), however, determined that neither conversion of propionate to glucose nor phosphoenolpyruvate carboxykinase mRNA abundance was affected by feed restriction. The relative contribution of propionate to glucose is reduced during periods of increased glucose demand, whereas the contribution of Ala and lactate becomes more important (Overton et al., 1999; Reynolds et al., 2003; Larsen and Kristensen, 2013).

In vivo measurements from cows in our experiment showed that neither serum glucose nor liver glycogen concentration was significantly affected by far-off treatment (Dann et al., 2006). Liver from a subset of cows fed 150NRC or 80NRC in this study was used to determine mRNA abundance for PC (Lor et al., 2006). Comparisons of 150NRC and 80NRC showed that abundance of PC mRNA was elevated around parturition for both treatments, but the increase was greater for cows fed 80NRC. The authors proposed that for cows fed restricted energy, higher PC mRNA was likely associated with the need to maintain blood glucose concentrations, whereas cows fed 150NRC had greater availability of propionate from ruminal fermentation and thus had greater hepatic glucose output (Lor et al., 2006).

Knapp et al. (1992) indicated that the ratio of glucose:CO₂ from metabolism of [1-¹⁴C] propionate provides an index of the theoretical efficiency of gluconeogenesis. Because of randomization of label from [1-¹⁴C] propionate in the tricarboxylic acid cycle, 1 mol of labeled glucose should be produced for every mole of labeled CO₂ produced in the conversion of oxaloacetate to phosphoenolpyruvate. This would yield a theoretical efficiency of 1 for the ratio of glucose to CO₂ if every molecule of labeled propionate that forms oxaloacetate

were shunted directly into the gluconeogenic pathway. On d -30, the ratio of glucose:CO₂ was numerically lower for liver slices from cows fed 150NRC compared with that for 100NRC or 80NRC (Table 4). On d -30, -14, 1, and 14 the ratio of glucose, CO₂ was numerically higher for 100NRC cows than for 150NRC and 80NRC cows, but the ratio increased postpartum for 80NRC cows (far-off treatment by day interaction, $P = 0.03$). Propionate supply to the liver might have declined with the lower fermentable energy of the 100NRC diet compared with the higher-energy diets fed during the far-off period. Reduced propionate supply to the liver may result in a reduction in conversion of propionate to CO₂ despite similar conversion to glucose. Although differences were not significant, these data suggest that propionate was directed toward gluconeogenesis at the expense of oxidation and that this effect was mediated by far-off dietary treatment. Odensten et al. (2005) observed a reduction in the molar proportion of propionate in the rumen due to lower DMI and diet digestibility in cows fed straw ad libitum. Intake was not depressed by feeding 100NRC in this study (Dann et al., 2006). Nozière et al. (2000) observed a decrease in propionate supply to the liver as DMI declined due to feed restriction. Neither conversion of propionate to glucose nor PEPCK mRNA abundance was affected by feed restriction (Velez and Donkin, 2005). The ratio was greater for CUA than for CUR at d 1 and 14 (Table 3; close-up treatment by day interaction, $P = 0.05$), indicating that greater nutrient supply immediately prepartum made gluconeogenesis more efficient.

Metabolism of propionate appears to be modulated during the transition period. Overton (1998) found that propionate conversion to glucose by liver slices was 19% and 29% greater at d 1 and 21 postpartum, respectively, than at d 21 prepartum. Aiello et al. (1984) found that the rates of gluconeogenesis from [1-¹⁴C] propionate in liver slices were 3-fold greater at d 30 of lactation than in slices from the same cows at 90 and 180 d of

lactation. McCarthy et al. (2015) found that cows treated with monensin had an increase in the ratio of glucose to CO₂, likely from an increase in propionate supply, indicating an increased propensity to convert propionate to glucose with increased propionate supply. Likewise, rates of glucose synthesis from propionate were greater in hepatocytes isolated from lactating goats than in those from nonlactating goats (Aiello and Armentano, 1987).

When the supply of propionate for hepatic gluconeogenesis is limiting, AA and lactate are used more readily as gluconeogenic substrates (Baird et al., 1980). Overton (1998) used Ala as a substrate for gluconeogenesis and found that Ala conversion to glucose at 1 and 21 d postpartum was 198 and 150% of that at 21 d prepartum. Greenfield et al. (2000) characterized temporal changes in mRNA abundance of PC and PEPCK around the time of calving. Abundance of PC mRNA increased 7.5-fold by the day after calving compared with that on d -28, reflecting adaptation to use of Ala and lactate as sources of glucogenic precursors. In contrast, changes in abundance of PEPCK mRNA lagged the increase observed for PC mRNA abundance such that PEPCK mRNA was unchanged on the day after calving but was elevated by d 28 postpartum (Greenfield et al., 2000). Alanine is released in the greatest amount of any AA by the portal-drained viscera and is removed by the liver in even greater amounts because of release of Ala by peripheral tissues involved in interorgan AA exchange (Bergman, 1986; Wolff et al., 1972; Aschenbach et al., 2010). Bell (1995) hypothesized that skeletal muscle serves as a labile pool of AA that is mobilized to support increased gluconeogenesis during the transition period. Calculations by Bell (1995) predicted that 500 g/d of endogenous protein is mobilized to meet glucose demands in a cow producing 30 kg of milk and consuming 15 kg DM/d during the first week of lactation. However, splanchnic flux data provide little evidence for marked upregulation of AA use for gluconeogenesis early postpartum, with the exception of Ala (Larsen and Kristensen, 2013).

Accumulation of TG in the liver may decrease the capacity of hepatocytes to perform gluconeogenesis, although conclusive evidence is lacking. Strang et al. (1998) showed that decreased ureagenesis was related to increased cellular TG accumulation in hepatocytes from ruminating calves. Exposure of hepatocytes from small ruminants to ammonia reduced uptake of propionate and resulted in lower rates of conversion of propionate to glucose (Demigné et al., 1986; Aiello and Armentano, 1987; Overton et al., 1999). Mills et al. (1986) observed a significant reduction in rates of gluconeogenesis from propionate, lactate, Ala, Asp, and Glu by liver slices from cows with induced fatty liver and ketosis. Bruss et al. (1986) fasted dry cows for 6 d and observed a reduced rate of propionate removal from blood, indi-

cating depressed liver function. Overfeeding during the dry period resulted in significantly reduced activity of PEPCK at 1 wk before and at 0.5 and 2 wk after parturition, suggesting that the rate of gluconeogenesis is decreased in cows with fatty liver (Rukkwamsuk et al., 1999b). In a subsequent experiment from the same group, Murondoti et al. (2004) overfed energy to cows during the dry period and then feed-restricted those cows postpartum. They observed a decrease in PC and PEPCK activity, suggesting a reduction in glucogenic capacity in cows with fatty liver. In our study, liver concentration of total lipid at 1 DIM was higher for cows with ad libitum intake than for cows with restricted intakes during the far-off or close-up periods (Dann et al., 2006); concentrations of TG followed a similar pattern, but differences were not statistically significant. Thus, cows in this study likely did not experience a degree of hepatic lipid infiltration that would be associated with reduced ability to perform gluconeogenesis.

Reduced ability to convert propionate to glucose by liver slices from 150NRC cows relative to 80NRC cows may indicate a lower enzymatic capacity to perform gluconeogenesis in cows allowed to overconsume energy during the far-off period. Insulin plays a central role in regulating blood glucose concentrations. Overfeeding energy during the dry period may result in prolonged high insulin concentrations that may lead to downregulation of key regulatory enzymes such as PEPCK and PC. In cows overfed energy throughout the dry period to induce fatty liver, hepatic activities of PEPCK were significantly reduced relative to controls at 1 wk before as well as 0.5 wk and 2 wk after parturition (Rukkwamsuk et al., 1999b). Similarly, Murondoti et al. (2005) also induced fatty liver in cows by overfeeding during the dry period followed by a short period of feed deprivation and measured lower activities of all gluconeogenic enzymes relative to cows that were fed more moderate energy diets. In contrast to overfeeding energy, 50% feed restriction in mid-lactation cows resulted in increased hepatic PC mRNA, but unchanged PEPCK mRNA, indicating the effect of dietary treatment on the capacity for gluconeogenesis from AA and lactate (Velez and Donkin, 2005).

Flux of carbon from propionate for gluconeogenesis, as regulated by PEPCK, occurs constantly to meet glucose requirements. Insulin affects the rate of incorporation of nonpropionate glucose precursors, such as Ala, into glucose through effects on PC. In sheep, infusion of insulin at low concentrations reduced the hepatic removal of lactate, Gln, and glycerol, and at higher concentrations reduced hepatic removal of Ala and pyruvate (Brockman, 1985). Additionally, overfeeding energy during the dry period may result in reduced insulin sensitivity in adipose tissue. Reduced insulin sensitivity in adipose may result in excessive NEFA mobilization and contribute to

TG accumulation in the liver, which may decrease the capacity of the liver to perform gluconeogenesis.

CONCLUSIONS

Prepartum nutrient intake altered the hepatic capacity for gluconeogenesis in periparturient cows. Overfeeding energy during the far-off dry period decreased the capacity of hepatic tissue for gluconeogenesis, whereas restricting feed intake had the opposite effect. Rates of conversion of Ala to glucose tended to be higher prepartum for cows fed restricted amounts versus those fed ad libitum. Similarly, use of propionate for gluconeogenesis was significantly greater prepartum for cows restricted in intake versus those fed ad libitum, was numerically greater on the day of calving, and continued to remain greater during early lactation. Feeding a close-up diet at either ad libitum or restricted intake had little effect on gluconeogenesis from either Ala or propionate. Allowing cows to overconsume energy relative to requirements during the far-off dry period decreased peripartal hepatic gluconeogenic capacity from both propionate and Ala. Controlling energy intake during the far-off dry period may confer an advantage to the cow by adapting metabolic processes before parturition, which may result in improved health and productivity.

NOTES

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Nonstandard abbreviations used: CU = close-up; CUA = close-up diet with ad libitum intake; CUR = close-up diet in restricted amounts; FO = far-off; NEFA = nonesterified fatty acids; PC = Pyruvate carboxylase; PEPCK = phosphoenolpyruvate carboxykinase; TCA = XXX; TG = triglycerides; 80 = moderate energy diet fed at restricted intake to meet 80% of NEL requirements; 100NRC = control diet; 150 = moderate energy diet fed ad libitum to exceed NRC recommendation for NEL by at least 50%.

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